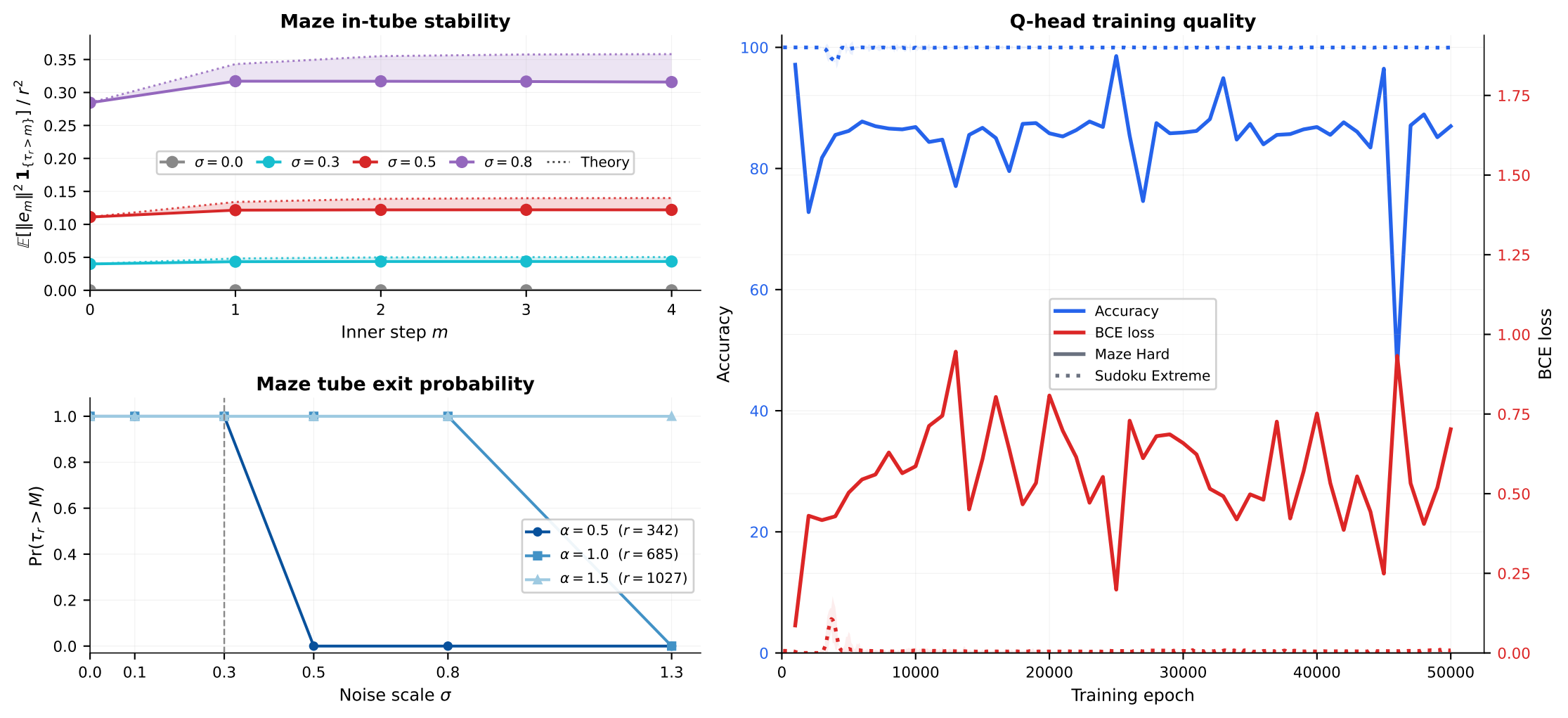
Maze in-tube stability
$\mathbb{E}[\|e_m\|^2 \mathbf{1}_{\{\tau_r > m\}}] / r^2$
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
$\sigma = 0.0$
$\sigma = 0.3$
$\sigma = 0.5$
$\sigma = 0.8$
Theory
0
1
2
3
4
Inner step m
Maze tube exit probability
$\Pr(\tau_r > M)$
1.0
0.8
0.6
0.4
0.2
0.0
$\alpha = 0.5$ $(r = 342)$
$\alpha = 1.0$ $(r = 685)$
$\alpha = 1.5$ $(r = 1027)$
0.0
0.1
0.3
0.5
0.8
1.3
Noise scale σ
Q-head training quality
Accuracy
100
80
60
40
20
0
BCE loss
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Accuracy
BCE loss
Maze Hard
Sudoku Extreme
0
10000
20000
30000
40000
50000
Training epoch